

Western Mathematics

2020

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks :
100**

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 – 27)

- Attempt Questions 11 – 27
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

1. What amount does an investment of \$20 000 grow to after 3 years at 5% p.a. compounded quarterly?
 - A. \$20 759.41
 - B. \$23 152.50
 - C. \$23 215.09
 - D. \$23 223.68

2. The weekly pay for workers at the Prosper Factory is normally distributed, with a mean of \$750 and a standard deviation of \$35.
What percentage of workers earn below \$680 a week?
 - A. 0.15%
 - B. 2.5%
 - C. 5%
 - D. 47.5%

3. The function $y = \ln x$ is transformed by first being dilated vertically by a scale factor of 3 and then translated horizontally 4 units to the left.
Find the equation of the transformed function.
 - A. $y = 3 \ln x^4$
 - B. $y = 4 \ln x^3$
 - C. $y = 3 \ln (x - 4)$
 - D. $y = 3 \ln (x + 4)$

4. For the series $2\pi, \pi, \frac{\pi}{2}, \dots$, calculate the exact value of the sum of the first 6 terms.
- A. $\frac{63\pi}{64}$
- B. $\frac{7\pi}{2}$
- C. $\frac{977\pi}{256}$
- D. $\frac{63\pi}{16}$
5. The 7th term of an arithmetic sequence is 45 and the 11th term is 77.
Find the first term (a) and the common difference (d).
- A. $a = -3$ and $d = 8$
- B. $a = 3$ and $d = 8$
- C. $a = 8$ and $d = -3$
- D. $a = 8$ and $d = 3$
6. Twenty students sit a Chemistry test and the mean of their scores is 78.
Two students sit the test late and their scores are 95 and 83.
What is the new mean for the Chemistry test?
- A. 79
- B. 80
- C. 83
- D. 89

7. What is the equation of the axis of symmetry of the quadratic function

$$y = -k(x + b)^2 + c ?$$

A. $x = 2ak$

B. $x = b$

C. $x = 2b$

D. $x = -b$

8. A function is given by $f(x) = \begin{cases} \frac{3x^2}{125} & 0 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$

If this function is a continuous probability distribution, what is the area under the curve?

A. -1

B. 0.5

C. 1

D. 2

9. Find the derivative of $e^{x \sin 3x}$.

A. $e^{3x \cos 3x}$

B. $e^{x \sin 3x}(\sin 3x + 3x \cos 3x)$

C. $e^{x \sin 3x}$

D. $e^{x \sin 3x}(\sin 3x - 3x \cos 3x)$

10. Holly drew a scatter-plot of a binomial data set which compared the construction time of houses with their cost.

The construction times ranged from 6 weeks to 6 months.

She found the equation of the line of best fit and used it to estimate the cost of a house which took 10 months to build.

What term would describe this process?

- A. Causality
- B. Correlation
- C. Extrapolation
- D. Interpolation

Western Mathematics

**2020 TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Class and Teacher

Mathematics Advanced

Student Number

Section II Answer Booklet

Student Name

90 marks

Attempt Questions 11 – 27

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (4 marks)

- (a) Show that the derivative of $\ln\left(\frac{3+x}{3-x}\right)$ is $\frac{6}{9-x^2}$.

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- (b) Hence or otherwise find $\int \frac{1}{9-x^2} dx$.

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Question 12 (5 marks)

5

Describe the features of the periodic function $y = 5 \cos [2(x - 3)] + 1$.

In your answer include the amplitude of the function, its period, the centre and the upper and lower endpoints of the vertical oscillation and the phase shift of the curve. You do not need to find x or y intercepts.

You may use a sketch to illustrate your answer if you wish, but it is not required.

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Question 13 (3 marks)

(a) Show that $\frac{4x - 3}{x} = \frac{-3}{x} + 4$

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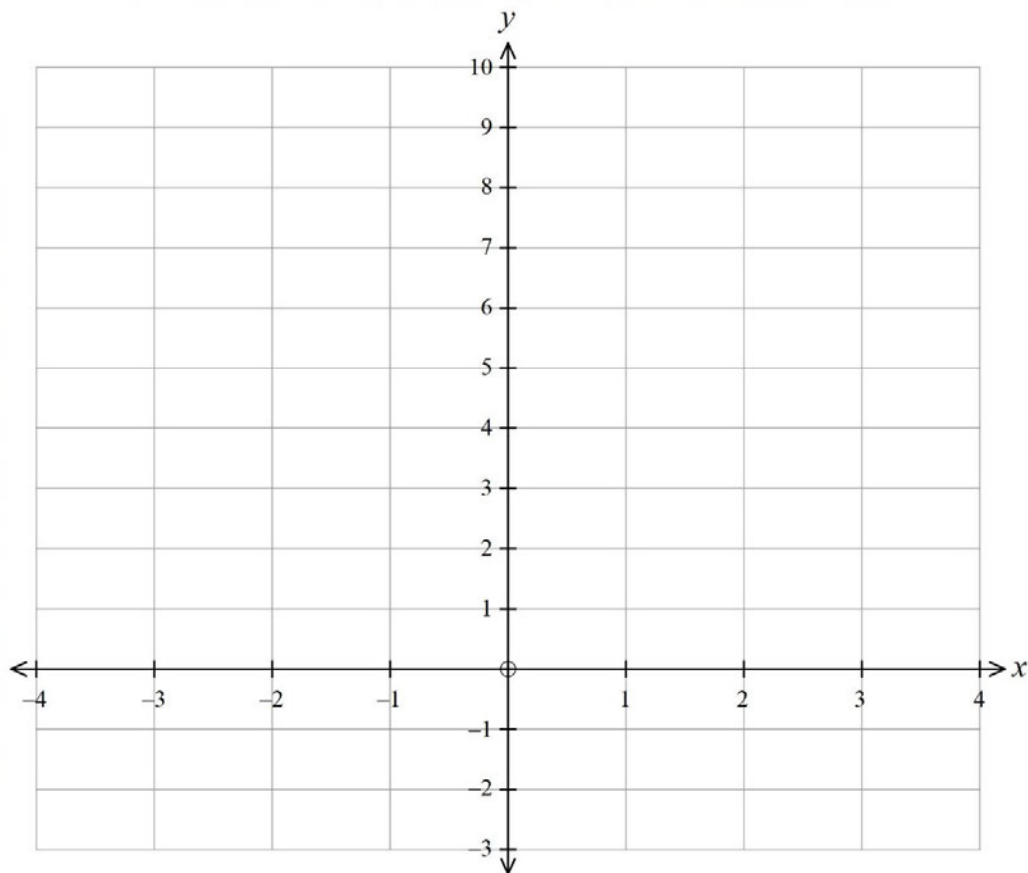
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(b) Hence or otherwise, sketch the graph of $y = \frac{4x - 3}{x}$ showing any asymptotes and the x -intercept.

2



Question 14 (5 marks)

Fred sits his Trial exams in Modern History and Ancient History.

The marks for the Modern History class have a mean of 54 and a standard deviation of 5.6.

The marks for the Ancient History class have a mean of 76 and a standard deviation of 2.1.

- (a) Compare and contrast the distribution of marks for the two classes. **2**

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- (b) Fred scored 65 for Modern History and 80 for Ancient History. **3**
Using z-score calculations, explain which subject he performed better in and why.

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Question 15 (6 marks)

(a) Solve $2 \cos \left[2 \left(x - \frac{\pi}{6} \right) \right] = 1$ in the domain $[0, 2\pi]$.

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(b) Find the median of the continuous probability distribution defined as $f(x) = \frac{x^3}{64}$ in the domain $[0, 4]$.

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Question 16 (7 marks)

- (a) Solve the equations $y = \sqrt{x+1}$ and $y = 11-x$ simultaneously and show that there is only point of intersection. Give its coordinates.

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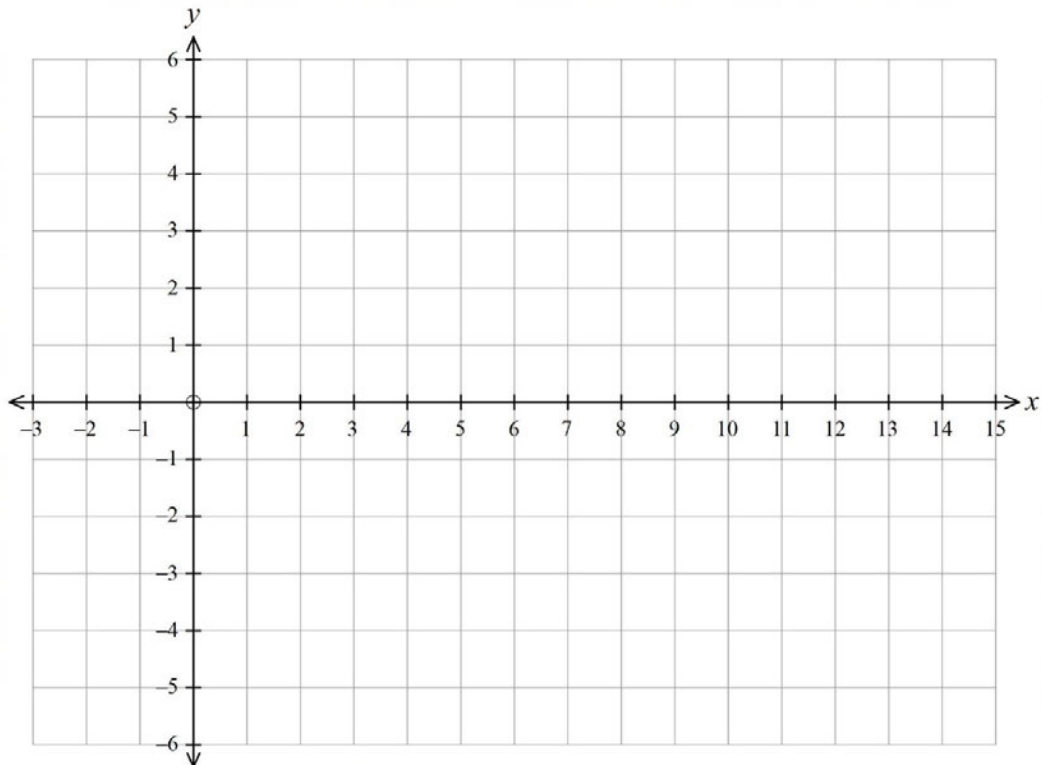
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- (b) Sketch $y = \sqrt{x+1}$ and $y = 11-x$ on the axes below.

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Question 16 continues on page 13

Question 16 continued

- (c) Calculate the area bounded by the curves $y = \sqrt{x+1}$ and $y = 11-x$ and the x -axis.

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Question 17 (8 marks)

It is known at the beginning of winter in a large population, 15% of people will be infected with a particular virus.

- (a) Four people are selected at random, find the probability that at least one of them has the virus. **2**

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- (b) What is the smallest number of people a drug company would need to test to have a greater than 95% chance that at least one of the tested people had the virus. **3**

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Question 17 continues on page 15

Question 17 continued

- (c) As winter progresses the virus spreads much more and the health authorities decide they want to stop the virus and have been given a new medication to trial. The two-way table shows the number of people in a trial.

	Taking Medication	Control Group
Virus	204	205
No Virus	212	209

1

- (i) What percentage of people in the trial had the virus?

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- (ii) What percentage of people in the control group had the virus?

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- (iii) Determine if it is worth the health authorities using this new medication.

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Question 18 (4 marks)

(a) Evaluate $\int_1^{e^3} \frac{5}{x} dx$.

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- (b) Henderson's harvests oranges to sell to Cottonworths, and the weights of the oranges they sell are normally distributed.

2

Oranges that weigh less than 100 grams are rejected, and this harvest season 97.5% of their oranges are accepted to sell.

Cottonworths also offers a bonus for premium oranges that are greater than 130 grams and 16% of this seasons harvest are classed as premium.

Find the mean and standard deviation of the weights of the oranges.

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Question 19 (6 marks)

Max did a survey of a group of people he knew about their age and how much they earn each week. The results are shown in the table below.

Age (years) (x)	18	45	28	15	32	68
Wage (\$/week) ($W$)	715	2350	1530	438	1690	1320

- (a) Using your calculator find (r) the correlation coefficient and explain what type and strength of correlation this data gives. **2**

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- (b) Using your calculator find the equation of the least-squares regression line in the form $W = Bx + A$ where A and B are integers. **1**

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- (c) Use your equation to estimate the earnings of a 50 year-old worker. **1**

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Question 19 continues on page 18

Question 19 continued

- (d) Could your equation from part(b) be used to make valid estimates for ages greater than 68 and less than 15 years?

2

Validate your response with calculations and or reasons.

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Question 20 (7 marks)

A swimming pool is to be emptied for maintenance. The quantity of water, Q litres, remaining in the pool at a time, t minutes after it starts to drain, is given by:

$$Q(t) = 2000(25 - t)^2, \quad t \geq 0.$$

- (a) At what rate (in litres/min) is the water being removed at any time (t)? **1**

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- (b) How long will it take to remove half of the water from the pool to the nearest minute? **2**

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- (c) At what time does the rate of flow of water from the pool reach 20 kL/minute? **2**

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- (d) Describe how the amount of water remaining in the pool changes as the pool empties. **2**
Include mention of how the rate itself changes.

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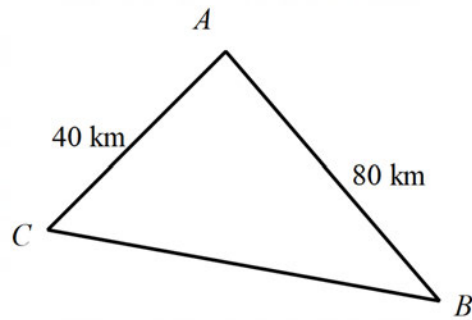
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Question 21 (4 marks)

Three towns, A , B and C form a triangle.

Town A is 80 km from Town B and Town C is 40 km from Town A as shown below:



The bearing of Town B from Town A is 130° . The bearing of Town C from Town A is 240° .

- (a) Find the area of the triangle formed by the three towns, to the nearest square kilometre. 2

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- (b) Using the cosine rule, find the distance between Town B and Town C , to the nearest kilometre. 2

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Question 22 (3 marks)

- (a) Given $f(x) = \sqrt{4 - x^2}$ complete this table of values, correct to 3 decimal places.

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x	0	0.5	1	1.5	2
$f(x)$					

- (b) Use the Trapezoidal rule, with four sub-intervals, to estimate the value of $\int_0^2 \sqrt{4 - x^2} dx$.

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Question 23 (6 marks)

For the curve $y = x^3 - 3x^2 - 9x + 4$:

- (a) Find any stationary points and classify them. **3**

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- (b) Find the point of inflexion. **1**

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- (c) Sketch the curve, showing all main features. **2**

Question 24 (7 marks)

A particle is moving in a straight line with velocity $v = 3e^t + 6e^{-t}$ with t measured in minutes and v in ms^{-1} .

The particle begins its motion at origin.

- (a) What is the initial velocity? 1

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- (b) Find an equation for x , the displacement of the particle. 2

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- (c) Show that when $x = 10$, $3e^{2t} - 7e^t - 6 = 0$. 2

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- (d) Hence, find the value of t when $x = 10$. 2

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Question 25 (5 marks)

Kate and Dave are buying a house for \$1 700 000, they have a \$200 000 deposit and will need to borrow the remaining balance.

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An interest rate of 3.6% p.a. compounded monthly is charged on the outstanding balance. The loan is to be repaid in equal monthly payments (M) over a 30 year period.

How much should Kate and Dave be paying each month to fully pay off the house in the 30 year time period and how much interest do they pay over the life of the loan?

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 26 (5 marks)

A leak from a tanker has accidentally contaminated a farmer's paddock with a toxic chemical.

The chemical concentration in the soil was 6kL/ha immediately after the accident.

One year later the concentration was measured to be 2.4kL/ha.

It is known that the concentration, C , is given by:

$$C = C_0 e^{-kt}$$

Where C_0 and k are constants and t is measured in years.

- (a) Evaluate C_0 and k .

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- (b) It will not be safe for the farmer to plant a new crop until the concentration falls below 0.2kL/ha.

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How long, to the nearest month, after the spill does the farmer need to wait for the paddock to be safe to use?

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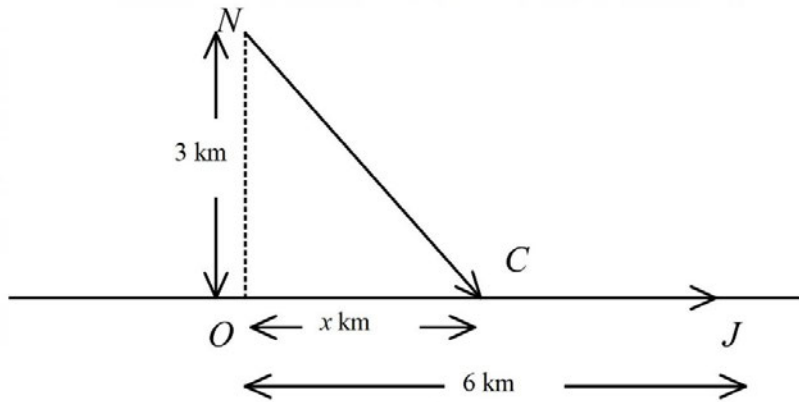
Question 27 (5 marks)

5

Nathan is on a paddle board in the ocean 3 kilometres from the nearest point O on a straight beach.

Nathan needs to meet his friend Jarrod who is located 6 kilometres along the beach from the point O .

Nathan paddles at a rate of 4 km/h to a point C on the beach and then walks at a rate of 5km/h along the beach to Jarrod.



Show that the total time it takes Nathan to reach Jarrod is given by:

$$T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}.$$

Hence, find the minimum time it will take Nathan to reach Jarrod.

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Answer Space for Question 27 continues on page 27

Question 27 continued

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End of Paper

Section II Extra writing space

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Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement**Length**

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

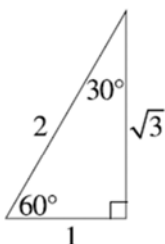
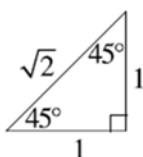
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

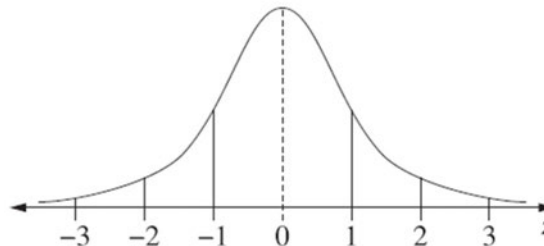
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^n C_x p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \left\{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = a + \lambda b$$

Complex Numbers

$$\begin{aligned} z &= a + ib = r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Western Mathematics
2020 Trial Higher School Certificate Examination
Mathematics Advanced

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

 A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

 A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

Western Mathematics

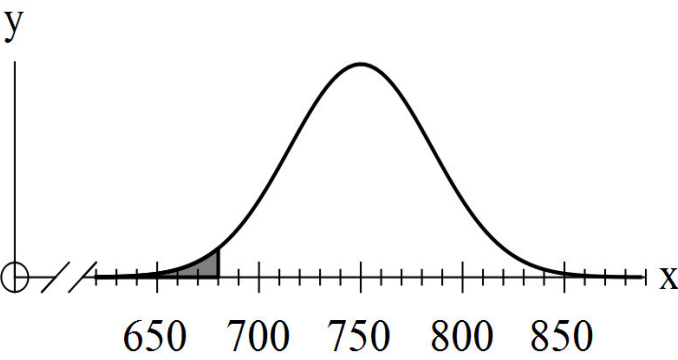
2020

TRIAL HSC EXAMINATION

Mathematics Advanced Solutions

Section I

No	Working	Answer
1.	$P = \$20000 \quad r = \frac{0.05}{4} \quad n = 12$ $P = 20000 \left(1 + \frac{0.05}{4} \right)^{12}$ $= \$23\,215.09$	C

No	Working	Answer
2.	 < \$680 means a z-score lower than -2 Therefore $100 - 50 - 47.5 = 2.5\%$ so 2.5% of people will earn below \$680.	B
3.	$y = \ln x$ $y = 3 \ln x \text{ vertical dilation of } 3$ $y = 3 \ln (x + 4) \text{ horizontal translation to left } 4 \text{ units}$	D
4.	This is a geometric series with common ratio $r = \frac{1}{2}$ and first term $a = 2\pi$. $S_n = \frac{a(1 - r^n)}{1 - r}$ $S_n = \frac{2\pi \left(1 - \left(\frac{1}{2} \right)^6 \right)}{1 - \frac{1}{2}}$ $= \frac{63\pi}{16}$	D

No	Working	Answer
5.	$T_n = a + (n-1)d$ $T_7 = 45 \quad T_n = a + 6d \quad \textcircled{1}$ $T_{11} = 77 \quad T_n = a + 10d \quad \textcircled{2}$ $\textcircled{2} - \textcircled{1}$ $4d = 32$ $d = 8 \quad \textcircled{3} \text{ sub into } \textcircled{1}$ $45 = a + 6 \times 8$ $a = -3$ $a = -3 \text{ and } d = 8$	A
6.	$78 = \frac{\Sigma x}{20}$ $\Sigma x = 78 \times 20$ $= 1560$ $\bar{x}_{\text{new}} = \frac{1560 + 95 + 83}{22}$ $\bar{x}_{\text{new}} = 79$	A
7.	Since $(x + b)$ parabola is translated b units to the left, so axis is $x = -b$ OR Axis of symmetry $x = -\frac{b}{2a}$ $y = -k(x + b)^2 + c$ $= -k(x^2 + 2xb + b^2) + c$ $= -kx^2 - 2kbx - kb^2 + c$ $x = \frac{-2kb}{-2k}$ $= -b$	D

No	Working	Answer
8.	Area under the curve must equal 1. $ \begin{aligned} A &= \int_0^5 \frac{3x^2}{125} dx \\ &= \frac{3}{125} \int_0^5 x^2 dx \\ &= \frac{3}{125} \left[\frac{x^3}{3} \right]_0^5 \\ &= \frac{3}{125} \times \frac{125}{3} - 0 \\ &= 1 \end{aligned} $	C
9.	$\frac{d}{dx}(e^{x \sin 3x}) = (e^{x \sin 3x}) \times \frac{d}{dx}(x \sin 3x)$ Find $\frac{d}{dx}(x \sin 3x)$ $ \begin{aligned} u &= x & v &= \sin 3x \\ u' &= 1 & v' &= 3 \cos 3x \\ vu' + uv' &= \sin 3x + 3x \cos 3x \end{aligned} $ $\frac{d}{dx}(e^{x \sin 3x}) = (e^{x \sin 3x}) (\sin 3x + 3x \cos 3x)$	B
10.	Predicting values outside the original data set is called extrapolation.	C

Western Mathematics
2020 Trial Higher School Certificate Examination
Mathematics Advanced

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 25 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

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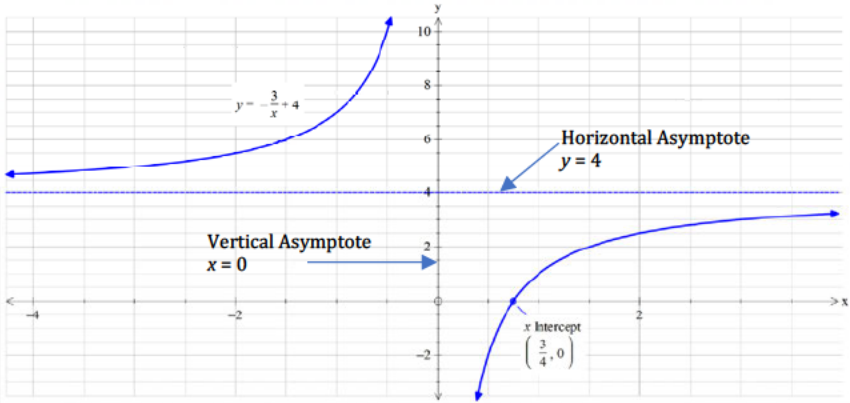
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|-----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 2. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 5. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☒ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☒ |
| 8. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |
| 9. | A | ☐ | B | ☒ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☒ | D | ☐ |

Western Mathematics Mathematics Advanced Trial HSC

Solutions 2020

Section II

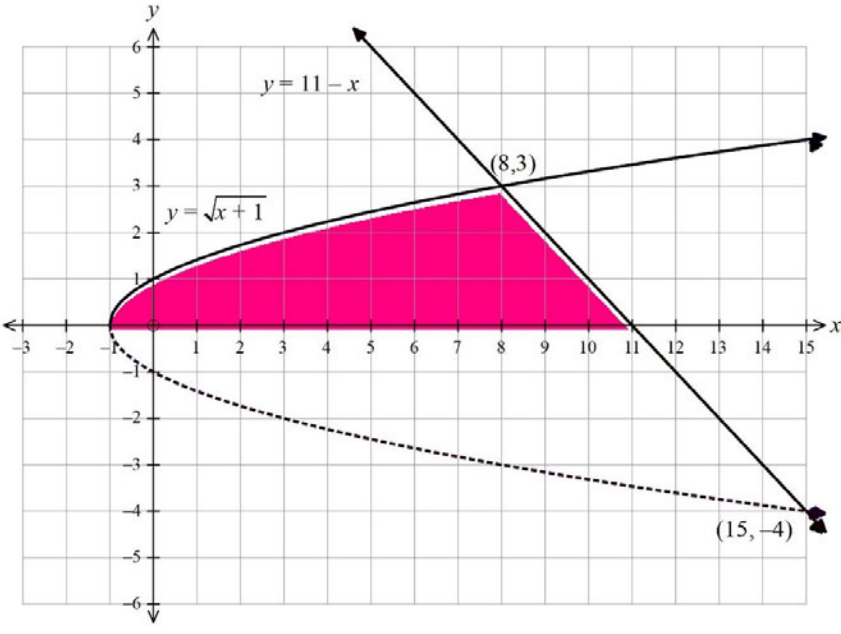
Question	Working and answer	Marks	Mark Allocation
11. a)	$y = \left[\ln \left(\frac{3+x}{3-x} \right) \right]$ $y = \ln(x+3) - \ln(3-x)$ $y' = \frac{1}{3+x} - \frac{-1}{3-x}$ $= \frac{3-x+3+x}{9-x^2}$ $= \frac{6}{9-x^2}$	2	2 marks for correct solution 1 mark for correct use of log laws or correct steps in differentiation or equivalent merit
	b) $\int \frac{1}{9-x^2} dx$ $= \frac{1}{6} \int \frac{6}{9-x^2} dx$ $= \frac{1}{6} \ln \left[\frac{3+x}{3-x} \right] + c$	2	2 marks for correct solution 1 for taking out the common factor to get part (a) answer or equal merit

Question	Working and answer	Marks	Mark Allocation
12.	a) $y = 5 \cos[2(x - 3)] + 1$ $y = k \cos[a(x + b)] + c$ Amplitude = $k = 5$ $\text{Period} = \frac{2\pi}{a} = \frac{2\pi}{2} = \pi$ Centre = 1 Min endpoint = $1 - 5 = -4$ Max endpoint = $1 + 5 = 6$ Phase - shift is 3 units to the right, due to $(x - 3)$	5	5 marks for all 5 features given. 1 mark for each correct feature.
13.	a) $\frac{4x - 3}{x} = \frac{4x}{x} - \frac{3}{x}$ $= 4 - \frac{3}{x}$ $= -\frac{3}{x} + 4$	1	1 for correct solution
	b) 	2	2 marks for diagram with correct shape, asymptotes and x-intercept 1 mark if shape, asymptotes or x-intercept are incorrect

Question		Working and answer	Marks	Mark Allocation
14.	a)	The Ancient History class has a higher mean and a smaller standard deviation, meaning the scores are grouped more closely around the mean which indicates more consistent marks. While the Modern History has a lower mean and a larger standard deviation, meaning the scores are more spread out around the lower mean, so are much less consistent marks.	2	2 if correctly commented on both mean and standard deviation 1 if only comment on either mean or standard deviation
	b)	$z = \frac{x - \mu}{\sigma}$ MH $x = 65$ $\sigma = 5.6$ $\mu = 54$ $z_{MH} = \frac{65 - 54}{5.6} = 1.96428571$ AH $x = 80$ $\sigma = 2.1$ $\mu = 76$ $z_{AH} = \frac{80 - 76}{2.1} = 1.9047619$ Fred performed better in MH as he had a higher positive z -score meaning he performed further above the mean, compared to AH, even though he had a lower raw mark for MH.	3	3 marks for two correct z-score calculations and analysis 2 marks for correct z scores but incorrect analysis or equivalent merit 1 for a correct z – score calculation or equivalent merit

Question	Working and answer	Marks	Mark Allocation
15.	a) $2 \cos \left[2 \left(x - \frac{\pi}{6} \right) \right] = 1$ $\cos \left[2 \left(x - \frac{\pi}{6} \right) \right] = \frac{1}{2}$ Let $2 \left(x - \frac{\pi}{6} \right) = \theta$ $\cos \theta = \frac{1}{2} \text{ and } \cos \text{ +ve 1st \& 4th quad}$ and in repetitions of these positions $\theta = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$ If $\theta = 2 \left(x - \frac{\pi}{6} \right)$ $2 \left(x - \frac{\pi}{6} \right) = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}$ $x - \frac{\pi}{6} = -\frac{\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{13\pi}{6}$ $x = 0, \frac{\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi, \cancel{\frac{7\pi}{3}}$ $x = 0, \frac{\pi}{3}, \pi, \frac{4\pi}{3}, 2\pi$	3	3 marks for all correct solutions with no extras 2 marks for most correct solutions with some missing or extras or equivalent correct working 1 mark for some relevant and correct working

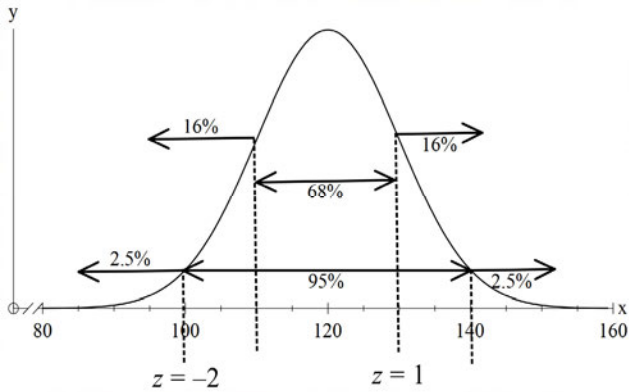
Question	Working and answer	Marks	Mark Allocation
	b) $f(x) = \frac{x^3}{64}$ Find CDF $\int_0^x \frac{x^3}{64} dx$ $= \frac{1}{64} \int_0^x x^3 dx$ $= \frac{1}{64} \left[\frac{x^4}{4} \right]_0^x$ $= \frac{1}{64} \left[\frac{x^4}{4} - \frac{0^4}{4} \right]$ $= \frac{1}{64} \left(\frac{x^4}{4} \right)$ $= \frac{x^4}{256}$ For Median $\int_0^x f(x) dx = 0.5$ $\frac{x^4}{256} = 0.5$ $x^4 = 128$ $x = 3.363585661$ $x = 3.4$ $P(X < 3.4) = 0.5$	3	3 marks for correct solution 2 marks for setting up integral correctly with error in calculation or integration or similar merit 1 for some attempt to set integral and evaluate

Question	Working and answer	Marks	Mark Allocation
16. a)	$\sqrt{x+1} = 11-x \quad \text{squaring both sides}$ $x+1 = (11-x)^2$ $x+1 = 121 - 22x + x^2$ $x^2 - 23x + 120 = 0$ $x^2 - 8x - 15x + 120 = 0$ $(x-8)(x-15) = 0$ $x = 8 \text{ or } 15$ When $x = 8$ $y = 3$ When $x = 15$ $y = -4$ Other methods of solution are possible. Since the radical sign implies positive square root, $y = \sqrt{x+1}$ only exists above the x axis, so the point $(15, -4)$ is not part of the solution The only point of intersection is $(8, 3)$.	3	3 marks for correct solution including eliminating the incorrect solution. 2 marks for giving both solutions with no explanation or equivalent merit 1 mark for some relevant working on solving the equations
b)		2	2 marks for two correct graphs (shading not required, nor is the dotted curve) 1 mark for only one graph correct or equivalent merit

Question	Working and answer	Marks	Mark Allocation
c)	$ \begin{aligned} A &= \int_{-1}^8 \sqrt{x+1} \, dx + \int_8^{11} 11-x \, dx \\ &= \int_{-1}^8 (x+1)^{\frac{1}{2}} \, dx + \int_8^{11} 11-x \, dx \\ &= \left[\frac{2(x+1)^{\frac{3}{2}}}{3} \right]_{-1}^8 + \left[11x - \frac{x^2}{2} \right]_8^{11} \\ &= \left[\frac{2(8+1)^{\frac{3}{2}}}{3} - \frac{2(-1+1)^{\frac{3}{2}}}{3} \right] + \left[\left(121 - \frac{121}{2} \right) - \left(88 - \frac{64}{2} \right) \right] \\ &= 18 + 4.5 \\ &= 22.5 \text{ units}^2 \end{aligned} $ Students could also find simple area of a triangle for the 2nd section of the integral $A = \frac{1}{2} \times 3 \times 3 = 4.5 \text{ units}^2$	2	2 marks for finding the required area by any valid method. 1 mark for finding the area under one of the curves or equivalent merit

17.	a)	$P(V) = 15\% \quad P(\bar{V}) = 85\%$ $P(\text{at least 1 virus in group of 4}) = 1 - P(\text{no virus in 4 people})$ $= 1 - P(\bar{V}\bar{V}\bar{V}\bar{V})$ $= 1 - (0.85)^4$ $= 0.47799375$ $= 0.478$	2	2 marks for the correct solution 1 mark for finding the complement or equivalent merit
	b)	Let V be the number of people with the virus in a group of size x $P(V \geq 1) = 1 - P(V = 0)$ $P(V = 0) = (0.85)^x$ Require $P(V \geq 1) > 0.95$ $1 - (0.85)^x > 0.95$ $1 - 0.95 > (0.85)^x$ $(0.85)^x < 0.05$ Take logs of both sides $\ln(0.85^x) < \ln 0.05$ $x \ln(0.85) < \ln 0.05$ [As $\ln(0.85) < 0$] $x > \frac{\ln(0.05)}{\ln(0.85)}$ $x > 18.4$ $\therefore$ 19 people would be the smallest number needing to be tested to find at least 1 person with the virus.	3	3 marks for the correct solution 2 marks for finding the correct equation to solve and attempting to solve it or equivalent merit 1 mark for some relevant and correct statements and calculations
	c)	(i) $\text{Total people} = 204 + 212 + 205 + 209 = 830$ $\text{Percentage of People with Virus} = \frac{204 + 205}{830} \times 100$ $= 49\%$	1	1 mark for correct answer
		(ii) Percentage of control group with virus = $\frac{205}{414} \times 100$ $= 49.5\%$ $\approx 50\%$	1	1 mark for correct answer

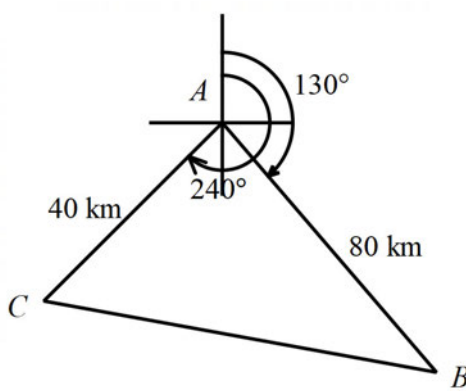
		(iii) Percentage of medication group with virus = $\frac{204}{416} \times 100$ $= 49.04\%$ $\approx 50\%$ It is not worth using the medication, since almost the same percentage of the virus occurs whether on medication or not.	1	1 mark for correct answer
18.	a)	$\int_1^{e^3} \frac{5}{x} dx = 5 \int_1^{e^3} \frac{1}{x} dx$ $= 5 [\ln x]_1^{e^3}$ $= 5(\ln e^3 - \ln 1)$ $= 5(3\ln e - 0)$ $= 15$	2	2 marks for correct solution 1 mark for correct integration with an error in evaluation or equivalent merit

	b) Students can draw a diagram or use calculations - any valid method can be used.  2.5 % of Oranges are rejected so since 5% lie outside the 2 sd limit 2.5% will be further than 2 standard deviations below the mean. Therefore 100 grams lies on the -2 z-score position. Since 16 % greater than 130 grams, similarly 16% will be at the equivalent position below (since distribution is symmetric), a total of 32 % outside these positions. Amount between these is $100\% - 32\% = 68\%$, so these correspond to the 1 standard deviation position. So the 16% premium Oranges are above 1 standard deviation above the mean. So between $z = -2$ (100) and $z = 1$ (130) is 3 z-scores. The standard deviation must therefore be $\sigma = \frac{130 - 100}{3} = 10$ The mean is one z score down from $z = 1$ so $\mu = 130 - 10 = 120$ $\therefore \mu = 120$ and $\sigma = 10$	2	2 marks for finding and stating the correct mean and standard deviation by any method 1 mark for finding one of the two values, or some correct working toward finding the two values

19.	a)	Scatterplot does not need to be drawn to answer the question, however can help as a visual aide. Using calculator $r = 0.5263217513$ $r = 0.53$ (2dp) This means a weak/medium positive correlation, due to the final data point being included.	2	2 marks for correct r value and for correct statement 1 mark for stating correct interpretation for incorrect r value or equivalent merit
	b)	$W = Bx + A$ From the calculator $B = 18.47948276 \quad \text{so } B = 18$ $A = 706.0377586 \quad \text{so } A = 706$ $\therefore W = 18x + 706$	1	1 mark for correct equation
	c)	$W = 18x + 706$ when $x = 50$ $W = 18 \times 50 + 706$ $W = \$1606$	1	1 mark for correct answer

	d)	The equation is not a valid way to extrapolate for younger and older workers, as it has only a moderate correlation coefficient. There is one data point (for a 68-year old) which affects the results causing this. People below 15 would not be working because they are too young, and at the older end, people have retired or are working less, and this equation does not take this into account. The equation is only valid for interpolation between 15 and 45 and, even then, has been affected by the last data point, a possible outlier. Max would need more data for ages outside of this if he is to make a more accurate prediction. Had he left the last data point out the correlation would have been much stronger and more accurate in the 15 – 45yr range.	2	2 marks for any valid response which shows understanding of correlation, linear regression and extrapolation

20.	a)	$Q(t) = 2000(25 - t)^2, t \geq 0$ $Q'(t) = -4000(25 - t)$ $\therefore$ it is emptying at a rate of $4000(25 - t)$ litres/minute	1	1 mark for correct answer
	b)	Pool full at $t = 0$ $Q(t) = 2\,000(25 - 0)^2$ $= 1\,250\,000 \text{ litres}$ $\therefore$ half full = 625 000 litres $625\,000 = 2\,000(25 - t)^2$ $312.5 = 625 - 50t + t^2$ $t^2 - 50t + 312.5 = 0$ $2t^2 - 100t + 625 = 0$ $t = \frac{-(-100) \pm \sqrt{100^2 - 4 \times 2 \times 625}}{2 \times 2}$ $t = \frac{100 \pm \sqrt{5000}}{4}$ $t = 7.322 \text{ or } 42.68$ $\therefore t = 7 \text{ minutes}$ $\therefore$ it will take ≈ 7 minutes to half empty the pool	2	2 marks for correct answer 1 mark for finding a quadratic equation and attempting to solve or equivalent merit
	c)	$20 \text{ kL} = 20\,000 \text{ L/min}$ $-20\,000 = -4\,000(25 - t)$ $-20\,000 = -100\,000 + 4\,000t$ $4\,000t = 80\,000$ $t = 20 \text{ min}$ $\therefore$ the flow rate will be 20kL after 20 minutes	2	2 marks for correct answer 1 for correct conversion from kL or correct calculation without conversion or similar merit
	d)	Pool is empty when $2000(25 - t)^2 = 0$ ie when $t = 25$ Rate = $Q'(t) = -4000(25 - t)$ When $t = 0$, Rate = $-4000 \times 25 = -100\,000$ When $t = 25$, Rate = $-4000 \times 0 = 0$ As the pool empties, the rate of flow remains negative, but its value is decreasing meaning that the rate of flow of water slows, becoming zero when the pool is empty	2	2 marks for any explanation that includes that the rate of flow is decreasing.

21.	a)	 $\angle CAB = 240^\circ - 130^\circ = 110^\circ$$A = \frac{1}{2} \times 40 \times 80 \times \sin 110^\circ$$= 1503.508193$$= 1504 \text{ km}^2$	2	2 marks for finding correct area 1 mark for finding the angle CAB from the bearings or equivalent merit												
	b)	$a^2 = 40^2 + 80^2 - 2 \times 40 \times 80 \cos 110^\circ$$= 10188.92892$$a = 100.9402245$$a = 101 \text{ km}$	2	2 marks for correct answer 1 mark for the correct substitution or equivalent merit												
22.	a)	<table><tr><td>x</td><td>0</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td></tr><tr><td>$f(x)$</td><td>2</td><td>1.936</td><td>1.732</td><td>1.323</td><td>0</td></tr></table>	x	0	0.5	1	1.5	2	$f(x)$	2	1.936	1.732	1.323	0	1	1 mark for correct answer
x	0	0.5	1	1.5	2											
$f(x)$	2	1.936	1.732	1.323	0											
	b)	$\int_a^b f(x)dx \approx \frac{b-a}{2n} [y_0 + y_n + 2 \times (y_1 + y_2 + \cdots y_{n-1})]$$\approx \frac{2}{8} [(2 + 0) + 2(1.936 + 1.732 + 1.323)]$$\approx 2.9955$	2	2 marks for correct estimate using formula or 4 trapezia 1 mark for some progress by any method												

23.	a)	$y = x^3 - 3x^2 - 9x + 4$ $y' = 3x^2 - 6x - 9$ $= 3(x^2 - 2x - 3)$ $= 3(x - 3)(x + 1)$ $y'' = 6x - 6$ $= 6(x - 1)$ Stat pts when $y' = 0$ $3(x - 3)(x + 1) = 0$ $x = 3$ or $x = -1$ when $x = 3$ $y = -23$ $y'' = 12 > 0 \therefore$ minimum when $x = -1$ $y = 9$ $y'' = -12 < 0 \therefore$ maximum	3	3 marks for correct points and (checked) nature 2 marks for finding y' and y'' and turning points but not classifying or equivalent merit 1 marks for finding y' and y'' or equivalent merit								
	b)	Inflexion when $y'' = 0$ $6(x - 1) = 0$ $x = 1$ when $x = 1$ $y = -7$ check concavity change <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td></tr><tr><td>y''</td><td>-6</td><td>0</td><td>6</td></tr></table> Concavity changes so (1, -7) is an inflexion	x	0	1	2	y''	-6	0	6	1	1 mark for correct point
x	0	1	2									
y''	-6	0	6									

	c)	Local Maximum (-1, 9) y Intercept (0, 4) Point of Inflection (1, -7) Local Minimum (3, -23) $y = x^3 - 3x^2 - 9x + 4$	2	2 marks for correct shape of the curve and the 4 points shown. 1 mark for an incorrect graph with some points correct
24.	a)	$v = 3e^t + 6e^{-t}$ when $t = 0$ $v = 3e^0 + 6e^0$ $= 9ms^{-1}$	1	1 mark for correct answer
	b)	$x = \int v dt$ $= \int 3e^t + 6e^{-t} dt$ $= 3e^t - 6e^{-t} + c$ When $t = 0 \quad x = 0$ $0 = 3 - 6 + c$ $c = 3$ $\therefore x = 3e^t - 6e^{-t} + 3$	2	2 marks for giving the correct equation 1 mark for correct integration but not finding C or equivalent merit

	c)	When $x = 10$ $10 = 3e^t - 6e^{-t} + 3$ $7 = 3e^t - 6e^{-t}$ $7 = 3e^t - \frac{6}{e^t}$ multiplying by e^t $7e^t = 3(e^t)^2 - 6$ $3e^{2t} - 7e^t - 6 = 0$	2	2 marks for correct substitution and manipulation to show required equation 1 mark correct substitution and some manipulation
	d)	Let $u = e^t$ $3u^2 - 7u - 6 = 0$ $3u^2 - 9u + 2u - 6 = 0$ $3u(u - 3) + 2(u - 3) = 0$ $(3u + 2)(u - 3) = 0$ $\therefore u = -\frac{2}{3}$ or 3 $e^t \neq -\frac{2}{3}$ does not exist $e^t = 3$ taking $\ln$ of both sides $t = \ln 3$ minutes	2	2 marks for substitution and solving a quadratic and finding 2 solutions 1 mark for substitution and attempt at solving the quadratic

25.	Amount borrowed = 1 700 000 – 200 000 = \$1 500 000 $P = \\$1\,500\,000$ $r = 3.6 \div 100 \div 12 = 0.003$ $n = 30 \times 12 = 360$ $A = P(1 + r)^n - M$ $A_1 = 1\,500\,000(1.003)^1 - M$ $A_2 = A_1(1.003)^1 - M$ $= 1\,500\,000(1.003)^2 - M(1.003) - M$ $A_3 = A_2(1.003)^1 - M$ $= 1\,500\,000[(1.003)^2 - M(1.003) - M](1.003) - M$ $= 1\,500\,000(1.003)^3 - M(1.003)^2 + M(1.003)^1 - M$ $= 1\,500\,000(1.003)^3 - M[1 + 1.003^1 + 1.003^2]$ Following this pattern $A_{360} = 1\,500\,000(1.003)^{360} - M[1 + 1.003^1 + 1.003^2 + \dots + 1.003^{359}]$ $A_{360} = 0$ as all money has been repaid $0 = 1\,500\,000(1.003)^{360} - M[1 + 1.003^1 + 1.003^2 + \dots + 1.003^{359}]$ $M = \frac{1\,500\,000(1.003)^{360}}{1 + 1.003^1 + 1.003^2 + \dots + 1.003^{359}}$ the denominator is a geometric series $a = 1 \quad r = 1.003 \quad n = 360$ $S_n = \frac{a(r^n - 1)}{r - 1}$ $S_{360} = \frac{1(1.003^{360} - 1)}{1.003 - 1}$ $\approx \$646.64$ $M = \frac{1\,500\,000(1.003)^{360}}{S_{360}}$ $= \$6819.68$ Total repaid = \$6819.68 $\times$ 360 = \$2 455 084.90 Interest Paid = \$2 455 084.90 – 1 500 000 = \$ 955 084.89	5	5 marks for correct repayment and amount of interest (NB allow for differences in calculations with rounding) 4 marks for correct repayment but not interest or equiv merit 3 marks for significant progress on series and equations or equiv merit 2 marks for setting up first 2 months or equiv merit 1 mark for some relevant calculations or equiv merit

26.	a)	$C = C_0 e^{-kt}$ At $t = 0$ $C = 6$ $6 = C_0 e^0$ $C_0 = 6 \text{ kL/ha}$ So $C = 6 e^{-kt}$ At $t = 1$ $C = 2.4$ $2.4 = 6 e^{-k(1)}$ $2.4 = 6 e^{-k}$ $0.4 = e^{-k} \quad (\text{taking ln of both sides})$ $k = -\ln(0.4)$ $k = 0.9162097319$ $k = 0.916$	3	3 marks for finding C_0 and writing a new equation for C and finding k 2 marks for finding C_0 writing a new equation for C and attempting to solve 1 mark for some relevant calculations and exponential manipulation
	b)	$C = 6 e^{-0.916t}$ If $C = 0.2$ $0.2 = 6 e^{-0.916t}$ $\frac{0.2}{6} = e^{-0.916t}$ $\frac{1}{30} = e^{-0.916t}$ $-0.916t = \ln\left[\frac{1}{30}\right]$ $t = \frac{\ln\left[\frac{1}{30}\right]}{-0.916}$ $t = 3.711919441$ $t = 3.7 \text{ years or 3 yrs 9 months}$	2	2 marks for finding t and stating how long the farmer should leave the field (Numbers could be different depending on rounding in working) 1 mark for showing some progress to being able to find t

27.	Using Pythagoras and $S = \frac{D}{T}$ Nathan paddles a distance of $\sqrt{x^2 + 9}$ at 4km/h $\text{Paddles} = \frac{\sqrt{x^2 + 9}}{4}$ Nathan walks a distance of $6 - x$ at 5km/h $\text{Walks} = \frac{6 - x}{5}$ Total time between Nathan and Jarrod $T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$ $T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$ $= \frac{1}{4}(x^2 + 9)^{\frac{1}{2}} + \frac{1}{5}(6 - x)$ $T'(x) = \frac{1}{8} \times 2x (x^2 + 9)^{-\frac{1}{2}} - \frac{1}{5}$ $= \frac{x}{4\sqrt{x^2 + 9}} - \frac{1}{5}$ $= \frac{5x - 4\sqrt{x^2 + 9}}{20\sqrt{x^2 + 9}}$ For minimum $T'(x) = 0$ $0 = \frac{5x - 4\sqrt{x^2 + 9}}{20\sqrt{x^2 + 9}}$ $0 = (5x - 4\sqrt{x^2 + 9})$ $5x = 4\sqrt{x^2 + 9} \quad (\text{square both sides})$ $25x^2 = 16x^2 + 144$ $9x^2 = 144$ $x^2 = 16$ $x = \pm 4 \quad x \neq -4$	5	5 marks for showing required expression and finding minimum 4 marks for showing required expression and minor error in finding minimum or equiv merit 3 marks for showing required expression some progress in finding minimum or equiv merit 2 marks for error in showing required expression and some progress in finding minimum or equiv merit 1 mark for some relevant calculations
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		check minimum when $x = 3$, $T'(x) < 0$ when $x = 5$, $T'(x) > 0$ $\therefore$ Min at $x = 4$ $T(x) = \frac{\sqrt{x^2 + 9}}{4} + \frac{6 - x}{5}$ $T(4) = \frac{\sqrt{4^2 + 9}}{4} + \frac{6 - 4}{5}$ = 1.65 hours = 1 hour & 39 mins		